

KARNATAKA BOARD

CLASS 10

SCIENCE

2019

Max. Marks: 80

General Instructions to the Candidate :

1. This Question Paper consists of 42 objective and subjective types of questions.
2. This question paper has been sealed by reverse jacket. You have to cut on the right side to open the paper at the time of commencement of the examination. Check whether all the pages of the question paper are intact.
3. Follow the instructions given against both the objective and subjective types of questions.
4. Figures in the right hand margin indicate maximum marks for the questions.
5. The maximum time to answer the paper is given at the top of the question paper. It includes 15 minutes for reading the question paper.

Four alternatives are given for each of the following questions / incomplete statements. Only one of them is correct or most appropriate. Choose the correct alternative and write the complete answer along with its letter of alphabet.

1. The change that occurs in the eye to see the distant objects clearly is

- (A) Focal length of the eye lens decreases
- (B) Curvature of the eye lens increases
- (C) Focal length of the eye lens increases
- (D) Culinary muscles of the eye contract

Ans;(C) focal length of the eye lens increases

2. The functional groups present in propanol and propanal respectively are

- (A) — OH and — CHO
- (B) — OH and — COOH
- (C) — CHO and — COOH
- (D) — CHO and — CO

Ans: (A) — OH and — CHO

3. The correct path of the movement of nerve impulses in the following diagram is

- (A) $Q \rightarrow S \rightarrow R \rightarrow P$
- (B) $P \rightarrow Q \rightarrow R \rightarrow S$
- (C) $S \rightarrow R \rightarrow Q \rightarrow P$
- (D) $P \rightarrow R \rightarrow S \rightarrow Q$

Ans: (D) $P \rightarrow R \rightarrow S \rightarrow Q$

4. The resistance of a conductor is $27\ \Omega$. If it is cut into three equal parts and connected in parallel, then its total resistance is

- (A) $6\ \Omega$ (B) $3\ \Omega$
- (C) $9\ \Omega$ (D) $27\ \Omega$

Ans: (B) $3\ \Omega$

5. The chemical equation that represents neutralization reaction among the following is

- (A) $\text{BaCl}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + 2\text{HCl}$
- (B) $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + 2\text{H}_2\text{O} + \text{Cl}_2$
- (C) $2\text{NaOH} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$
- (D) $\text{AgNO}_3 + \text{HCl} \rightarrow \text{AgCl} + \text{HNO}_3$

Ans: (C) $2\text{NaOH} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$

6. By constructing Khadin check-dams in level terrains,

- (A) Underground water level decreases

- (B) Underground water level increases
- (C) Vegetation in the nearby areas are destroyed due to excess moisture
- (D) Underground water gets polluted

7. To obtain a diminished image of an object from a concave mirror, position of the object should be

(F = principal focus, C = centre of curvature, P = pole)

- (A) Between C and F (B) beyond C
- (C) Between P and F (D) at F

8. The electronic configuration of element X is 2, 8, 8, 1 and the electronic configuration of element Y is 2, 8, 7. Then the type of bond formed between these two elements is

- (A) Covalent bond
- (B) Hydrogen bond
- (C) Metallic bond
- (D) Ionic bond

Ans; (D) ionic bond

9. Part of the flower that develops into fruit and part of the seed that develops into root respectively are

- (A) Ovary and plumule
- (B) plumule and radicle
- (C) Ovary and radicle
- (D) Ovary and ovule

Ans:(A) ovary and plumule

10. A pure dominant pea plant producing round — yellow seeds is crossed with pure recessive pea plant producing wrinkled — green seeds. The number of plants bearing round — green seeds in the 1 F generation of Mendel's experiment is

- (A) 0 (B) 1
- (C) 3 (D) 9

Ans:(A) 0

11. The functions of hormones are given in Column-A and the names of the hormones are given in Column-B. Match them and write the answer along with its letters: $4 \times 1 = 4$

Column - A	Column - B
(A) Prepares the body to deal with the situation	(i) Growth hormone
(B) Regulates metabolism for body growth	(ii) Testosterone
(C) Regulates blood sugar levels	(iii) Adrenaline
(D) Regulates the growth and development of the body	(iv) Progesterone
	(v) Insulin
	(vi) Thyroxine
	(vii) Oestrogen

Ans:

Column - A	Column - B
(A) Prepares the body to deal with the situation	Adrenaline
(B) Regulates metabolism for body growth	Thyroxine
(C) Regulates blood sugar levels	Insulin
(D) Regulates the growth and development of the body	Growth Hormone

Answer the following questions.

$7 \times 1 = 7$

12. Name the acid present in the stinging hair of nettle leaves.

Ans: Methanoic acid.

13. What are fossils?

Ans: Fossils are the remains or traces of ancient life that have been preserved by natural processes, from spectacular skeletons to tiny sea shells. Imprints, tracks and trails can also become fossilised, like dinosaur footprints or worm burrows

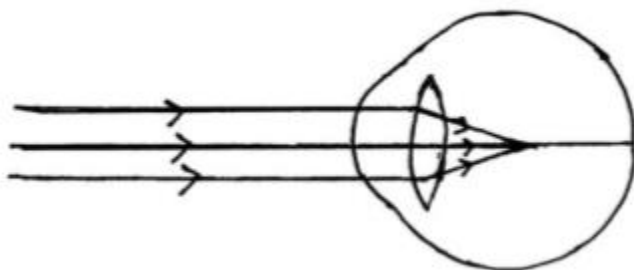
14. Convex mirror is commonly used as rear-view mirror in vehicles. Why?

Ans: Convex mirror is commonly used as rear-view mirror in vehicles because they always give an erect image and have field of view as their curved outwards.

15. What is roasting in metallurgy?

Ans: Heating of metallic ores strongly in the presence of excess air is called metallurgy.

16. Observe the given figure. Name the eye defect indicated in the figure and also mention the lens used to correct this defect.



Ans: Defect is Myopia or short Sightedness. It can be corrected by using concave lens of appropriate power.

17. What is Tyndall effect?

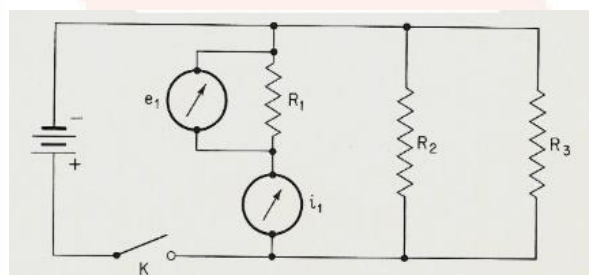
Ans: The Tyndall effect, also known as Tyndall scattering, is light scattering by particles in a colloid or particles in a fine suspension

18. Under what condition lactic acid is produced in the muscle cells?

Ans: Under anaerobic condition lactic acid is produced in the muscle cells.

Answer the following questions. $16 \times 2 = 32$.

19. Draw the diagram of an electric circuit in which the resistors $1 R$, $2 R$ and $3 R$ are connected in parallel including an ammeter and a voltmeter and mark the direction of the current.



20. Name the brown fumes liberated when lead nitrate is heated. Write the balanced chemical equation for this reaction.

Ans: The brown fumes liberated when lead nitrate is heated are of Nitrogen dioxide. This is the thermal decomposition reaction. The balanced chemical equation for this reaction is following:

$1\text{Pb}(\text{NO}_3)_2 \rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$ Thus, this will be the reaction when lead nitrate is heated and will release Nitrogen dioxide. You can further have a group discussion on the same.

21. Explain the process of translocation of food materials in plants.

Ans: Translocation occurs through part of vascular tissue known as phloem that also transports amino acids and other substances such as plant hormone which are made in tips of roots and shoots to the storage and growing organs.

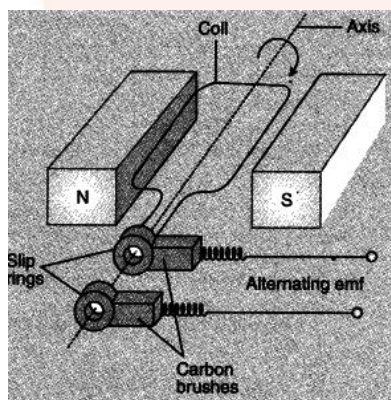
OR

Explain the process of digestion in the small intestine of man.

Ans: Small intestine is the site of completion of digestion of food into different components. Secretions from liver and pancreas enter the intestine to help in digestion process. The inner lining of small intestine has numerous finger-like projections called villi which increase the surface area for absorption. Wall of small intestine contains numerous glands that secrete intestinal juices containing amylolytic, proteolytic, lipolytic enzymes for the complete digestion of Carbohydrate, protein and fat.

22. Draw the diagram of a simple electric motor. Label the following parts:

(i) Split rings (ii) Brushes.



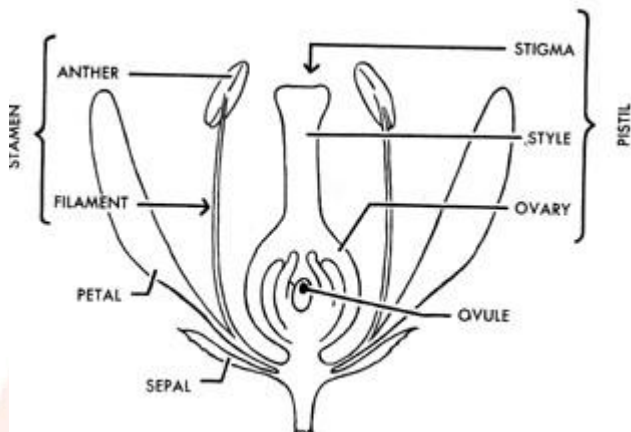
23. What are structural isomers? Name the first member of alkanes that shows structural isomerism.

Ans: Compounds with identical molecular formula but different structures are called structural isomers. 1 Butane or C_4H_{10}

24. Draw the diagram showing the longitudinal section of a flower.

Label the following parts:

(i) Style (ii) Anther.



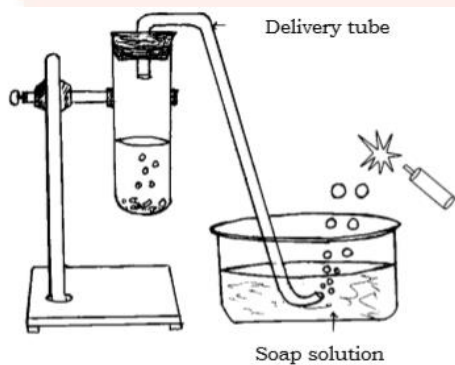
25. Draw the diagram of arrangement of apparatus used to show the reaction of zinc granules with dilute sulphuric acid and testing hydrogen gas by burning.

Label the following parts.

(i) Soap solution

(ii) Delivery tube.

Ans:



26. It is advantageous to connect electric devices in parallel instead of connecting them in series. Why?

Ans: Advantages of connecting electrical devices in parallel with the battery are as follows:

- (1) Voltage across each electrical device is same and the device can take current as per its resistance.
- (2) Total resistance in parallel circuit decreases, hence, a greater current may be drawn from the cell.

In a series circuit, if one component fails, the circuit is broken and none of the components work.

Each electrical appliance can be operated separately

OR

According to Joule's law of heating, mention the factors on which heat produced in a resistor depends. According to this law write the formula used to calculate the heat produced.

Ans: It states that the heat produced in a resistor is

- a) It is directly proportional to square of current, $H \propto I^2$
- b) It is directly proportional to resistance for a given current, $H \propto R$
- c) It is directly proportional to time for which current flows through the conductor, $H \propto t$.

So, $H = I^2 RT$

27. List the disadvantages of using fossil fuels.

Ans: The disadvantages of fossil fuel are following.

- The fossil fuels are non- renewable sources of energy.
- Burning of fossil fuel causes air pollution.
- The fossil fuel reserve in earth is limited and may get exhausted soon.

OR

List the advantages of 'reduce' and 'reuse' to save environment.

Ans: It is useful for people because many people gets benefit from items which are reused and recycled. These are cheaper methods. They need less space. We should minimize the use of various product which are obtained directly or indirectly from natural resources. We can save electricity by switching off fans and lights.

28. The focal length of a concave lens is 30 cm. At what distance should the object be placed from the lens so that it forms an image at 20 cm from the lens?

Focal length = 30 cm

$$1/f = 1/v - 1/u$$

$$1/30 = 1/20 - 1/u$$

$$1/u = 1/20 - 1/30$$

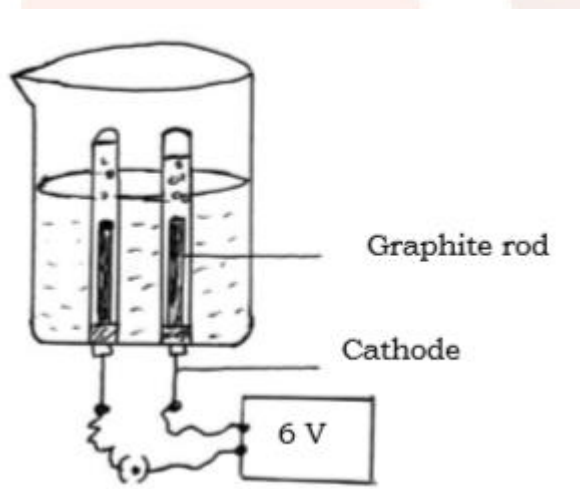
$$= 1/60 = 1/60$$

∴ the object should be placed from a distance of 60 cm.

29. Draw the diagram of the apparatus used in the electrolysis of water. Label the following parts.

(i) Graphite rod

(ii) Cathode.



30. Growth of thread like structures along with the gradual spoilage of tomato can be observed when a cut tomato is kept aside for four days. Interpret the causes for this change.

Ans: The changes occur in cut tomatoes kept in the open as the nutrients present in cut tomatoes provide a suitable medium for the growth of a pathogenic microorganism, especially Salmonella.

31. An electric refrigerator rated 400 W is used for 8 hours a day. An electric iron box rated 750 W is used for 2 hours a day. Calculate the cost of using these appliances for 30 days, if the cost of 1 kWh is Rs. 3/-.

Ans: Power = 0.4 KW

Energy = $0.4/8 = 0.05$ J

Cost for 1 KWH = Rs 3

Cost for 0.05 = 0.15 rs

For 30 days = 4.5 Rs

Power = 0.75 KWH

$E = P/T$

$E = 0.750 / 2$

= 0.375

Cost for 1 KWH = Rs 3

Cost for 0.375 = Rs 33.75 Rs Total Cost = $33.75 + 4.5 = 38.25$ Rs

32. There is no change in the colour of red litmus and blue litmus paper when introduced into an aqueous solution of sodium chloride. After passing direct current through the same solution, red litmus changes to blue colour. Which product is responsible for this change? Mention any two uses of this product.

Ans: Sodium Hydroxide is responsible for this change. Use of this product is:

- Sodium hydroxide is used in drain and oven cleaners.
- It is used in chemical manufacturing, oil refining, hydraulic fracturing water treatment and metal processing.
- It is also commonly used in the manufacture of fabric, plastic wrap, paper and soap.

33. A food chain in a polluted aquatic ecosystem is given. Observe it and answer the following questions.

Fresh water → Algae → Fishes → Birds.

- (i) **Which organisms are disturbed more due to bio magnification? Why?**

Ans: Algae is disturbed more due to bio magnification because it do the process of photosynthesis and consume more oxygen present in water.

(ii) This ecosystem will be destroyed gradually due to bio magnification. Why?

Ans: Bio magnification is a phenomenon that refers to progressive increase in the concentration of toxicants in organism at each trophic level. Harmful Chemicals and pesticides are sprinkled in the crop plants to protect them from pests and diseases. These Chemicals either wash down into soil or into water bodies. In this manner toxic Chemicals enter the food chain.

OR

A student places a piece of cucumber, a glass piece, a banana peel and a plastic pen in a pit and closes it. What changes can be observed in these materials after a month? Give scientific reason for these changes.

Ans: If a student places a piece of cucumber, a glass, a banana peel and a plastic pen in a pit and closes it then a piece of cucumber and banana peel will disappear because they are biodegradable materials, And plastic pen and a glass will remain as it was as they are non- biodegradable.

34. What is dispersion of light? Mention the colour that bends the least and the colour that bends the most when light undergoes dispersion through a prism.

Ans: The phenomenon of splitting of white light into its constituent colours when it passes through a Prism is called dispersion. This band of seven colours so obtained is VIBGYOR. V is for violet I is Indigo, B is blue G is green Y is yellow, O is orange and R is red. This is called spectrum.

OR

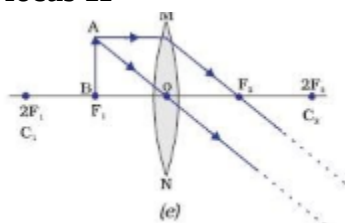
Mention any four phenomena that can be observed due to atmospheric refraction of light on the earth.

Ans: Some phenomena like twinkling of stars, early Sunrise, delayed sunset, stars seem higher than they actually are occurs due to atmospheric refraction.

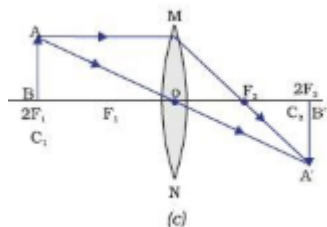
Answer the following questions. $5 \times 3 = 15$

35. Draw the ray diagrams for the image formation in a convex lens when an object is placed

(i) at focus $1F$



(ii) Beyond $2F$.

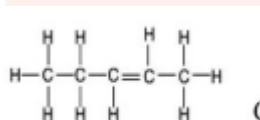


36. (i) Write the differences between saturated and unsaturated hydrocarbons.

Ans: The hydrocarbons in which all the carbon atoms are linked by only single bond are called saturated hydrocarbons. These compounds are normally not very reactive. Generally shows substitution reaction.

Hydrocarbons in which at least one double triple bond is present along with single bond is called unsaturated Hydrocarbons. These compounds are normally more reactive than saturated hydrocarbons. Generally shows addition reaction.

(ii) Write the molecular formula and structural formula of an alkene having five carbon atoms.



OR

(i) Carbon atom does not form $C4^-$ anion and $C4^+$ cation. Why?

Ans: Hence carbon cannot form $C4^-$ ions so easily. If carbon loses 4 electrons from its outer shell, it has to form C^{4+} ion. ... Carbon has to satisfy tetravalency, by sharing electrons with other atoms. so it has to form 4 covalent bonds either with its own atoms or atoms of other elements.

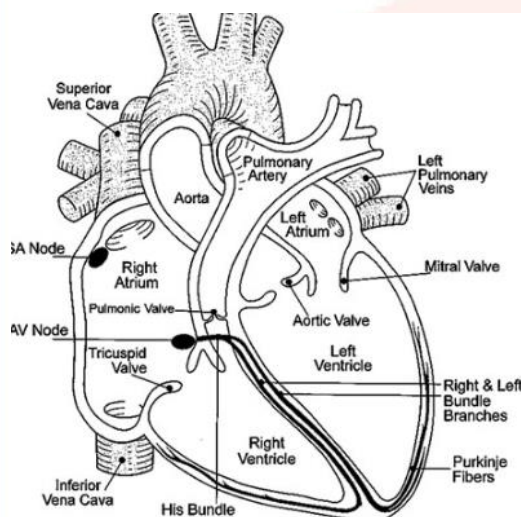
(ii) How can ethanol be converted into ethanoic acid?

Ans: The alcohol is oxidised by using potassium dichromate solution in the presence of dilute sulphuric acid. It is dichromate Ion that is the opposite oxidizing agent and dilute sulphuric acid is used to acidify the solution.

37. Draw the diagram showing the sectional view of the human heart. Label the following parts.

(i) Aorta

(ii) Chamber of the heart that receives deoxygenated blood.



Human Heart.

38. (i) Name the major constituent of biogas. Write the properties of biogas which make it a good fuel.

Ans: The main constituents of biogas are methane and CO_2 .

The advantages of biogas are followings:

- It burns without smoke therefore causes less pollution.
- Its calorific value is high and also used for lighting.

(ii) Name the two devices that work using heat energy of the sun.

Ans: Solar cooker and Satellite.

OR

(i) Write the advantages of solar cells.

- Solar cells create savings in electricity.
- It is a renewable source of energy.
- It is an environmental friendly energy. No pollution occurs.

(ii) Write any two hazards of nuclear power generation.

Ans: Two hazardous of nuclear power generation are:

- Contamination: Improper nuclear-waste storage and disposal result in environmental contamination.
- High cost of installation -Nuclear power plant needs a lot of money for their set up. Moreover the limited availability of uranium adds to the disadvantage.

39. Observe the given table and answer the following question:

Element	A	B	C	D	E
Atomic Number	11	4	2	7	19

Identify the two elements that belong to the same period and the two elements .That belong to the same group. Give reason for your conclusion.

Ans: Element B and D are in same period because their atoms have two shell.

Element A and element E are in same group because their outermost shell has one electron.

Answer the following questions. $3 \times 4 = 12$

40. (i) How does overload and short-circuit occur in an electric circuit ? Explain.

Ans: Short circuits happens because two electrical contacts touch over loading happens because too much resistance is placed on a circuit.

What is the function of fuse during this situation?

Ans: Fuses and circuit breaker are installed on the power out wire to trip or open to prevent the electrical current carrying wire from getting hot enough to melt the conductor insulation.

(ii) Mention two properties of magnetic field lines.

Ans: Two properties of magnetic field lines are followings:

The magnetic field lines originate from North Pole of the magnet and ends in its south pole by convention.

The magnetic field lines are closed in and continuous curve

41. Give reason:

(i) Ionic compounds in solid state do not conduct electricity, whereas in molten state are good conductors of electricity.

Ans: when Ionic are in a solid state, the ions cannot move and thus there is no flow of charge however when molten, the anions and cations are free to move therefore it is current flows and compound conducts electricity.

(ii) Silver articles when exposed to air gradually turn blackish.

Ans: Silver article when exposed to air gradually turn blackish because silver metal react with sulphur present in atmosphere and forms a layer of silver sulphide which is of black in colour.

(iii) Chemical reaction does not take place when copper is added to iron sulphate solution.

Ans: Copper being less reactive than iron, is placed at the bottom of the reactivity series, below iron. Therefore, a displacement reaction does not take place as copper cannot displace iron from iron sulphate solution.

OR

Give reason:

(i) “Alloys of iron are more useful when compared to pure iron.”

Ans: Alloying is a process of mixing of other metals to a particular metal so that it will be stronger. Iron is very weak in its pure state so why iron is alloyed with steel to make it strong so alloy of iron is strong than pure iron.

(ii) Copper loses its brown layer gradually when exposed to air.

Ans: Copper metal when exposed to air turns green in colour due to corrosion. When copper vessel is exposed to air in rainy season, the metal reacts with gases and moisture and atmospheric gases to form a mixture of copper carbonate and copper hydroxide. This gives a green colour to the surface of copper metal.

(iii) Aluminium oxide is called amphoteric oxide.

Ans: Metal oxides which react with both acids and as well as bases to produce salt and water are called amphoteric oxides. Aluminium oxide react both with bases and acids.

42. (i) Write the differences between homologous organs and analogous organs.

Ans: The organ which have the same functional structural but are different in function are homologous organs. For example Whale, bat, Cheetah and human share similarities in pattern of bones of forelimbs In these animals the same structure develop along different direction due to adaptation to different environment adaptation.

Analogous organs- These organs have similar functions but are different in there is structural details and origin .for example Wings of butterfly and birds are not Anatomical structure similar do they perform similar function. Analogous structures are a result of convergent evolution.

(ii) Write the differences between the sex chromosomes of man and sex chromosomes of woman.

Ans: In men and women 46 chromosomes are present .Out of these 2 are sex chromosome .The sex chromosome of female XX and in males are XY.

(iii) Sex of a child is determined by the father. How?

Ans: In human males two dissimilar chromosomes at present longer 'X' and shorter 'Y'. In human females two similar long 'X' chromosomes are present .All children's inherit 'X' chromosome from their mother. If the offspring inherit 'X' chromosome from the father then it will be a daughter and if it inherit 'Y' chromosome from the father then it will be a Son .Thus in human beings Sex of a child is determined by the father.